LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data

Student: _____ Date: _____

1. Objective

The objective is to predict the diabetes outcome from patient-related measurements. The target variable is **Outcome**: 0 indicates no diabetes and 1 indicates diabetes. This is an educational classification project and is not intended for medical diagnosis.

2. Dataset

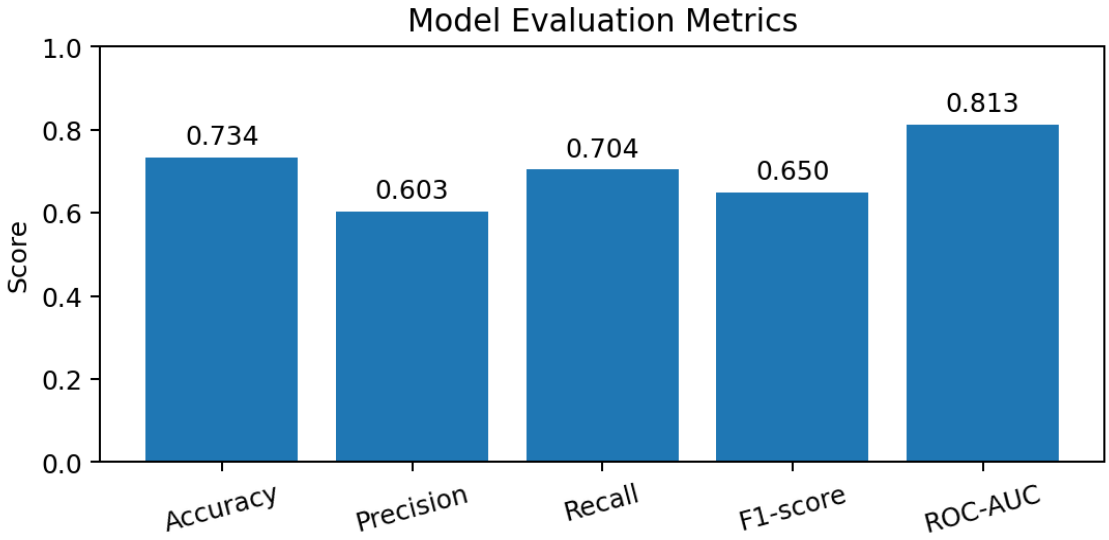
The project uses the Pima Indians Diabetes Database from Kaggle (file: **diabetes.csv**). It contains eight input features: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age. The target column is Outcome.

3. Model and Workflow

Stage	Description
Data inspection	Load the Kaggle CSV and inspect rows, columns, and data types.
Preprocessing	Replace invalid zero values with missing values; use median imputation and StandardScaler.
Data split	Split data into training and testing sets.
Model	Train balanced Logistic Regression.
Evaluation	Evaluate using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

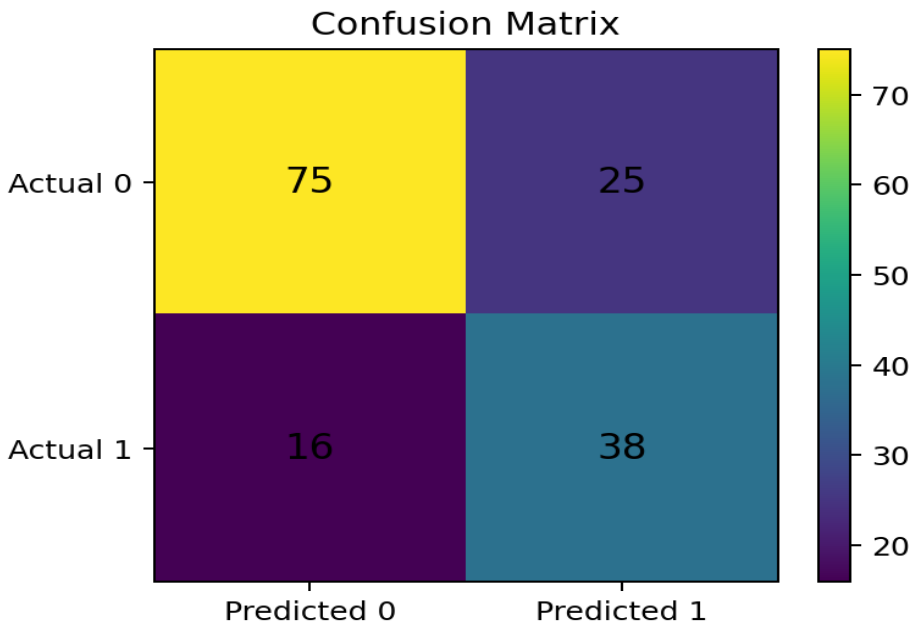
4. Evaluation Results

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.813	Class-separation ability


Model Evaluation Metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.734
0.603
0.704
0.650
0.813
Accuracy
Precision
Recall
F1-score
ROC-AUC

5. Confusion Matrix

The confusion matrix summarizes correct and incorrect predictions on the test set.



	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

6. Key Findings

The model obtains a ROC-AUC of approximately 0.813 and detects about 70.4% of positive cases in the fixed test split. Glucose and BMI have the strongest positive model coefficients according to the supplied project report. Recall is important in a screening-style classification task because false negatives represent positive cases that the model did not identify.

7. Limitations

The dataset is small and historical and represents a specific population. The model may not generalize to other populations or clinical settings. It is suitable for learning machine-learning classification concepts only and must not be used to diagnose diabetes.

8. Conclusion

This project demonstrates how Logistic Regression can be used for a binary classification task. The dataset is inspected and preprocessed, the model is trained and evaluated, and performance is measured using standard classification metrics. The results show useful class-separation ability, while also highlighting the need for careful interpretation and further validation.

Submission Checklist

- Add your name and date.
- Submit the executed .ipynb notebook and this PDF report.
- Keep metric outputs and graphs visible in the notebook.
- Write or personalize the final conclusion in your own words.

Note: The metric values and confusion-matrix counts in this report are reproduced from the reference project PDF supplied in this chat. Ensure the final executed notebook uses the same preprocessing workflow before claiming these results as its own output.